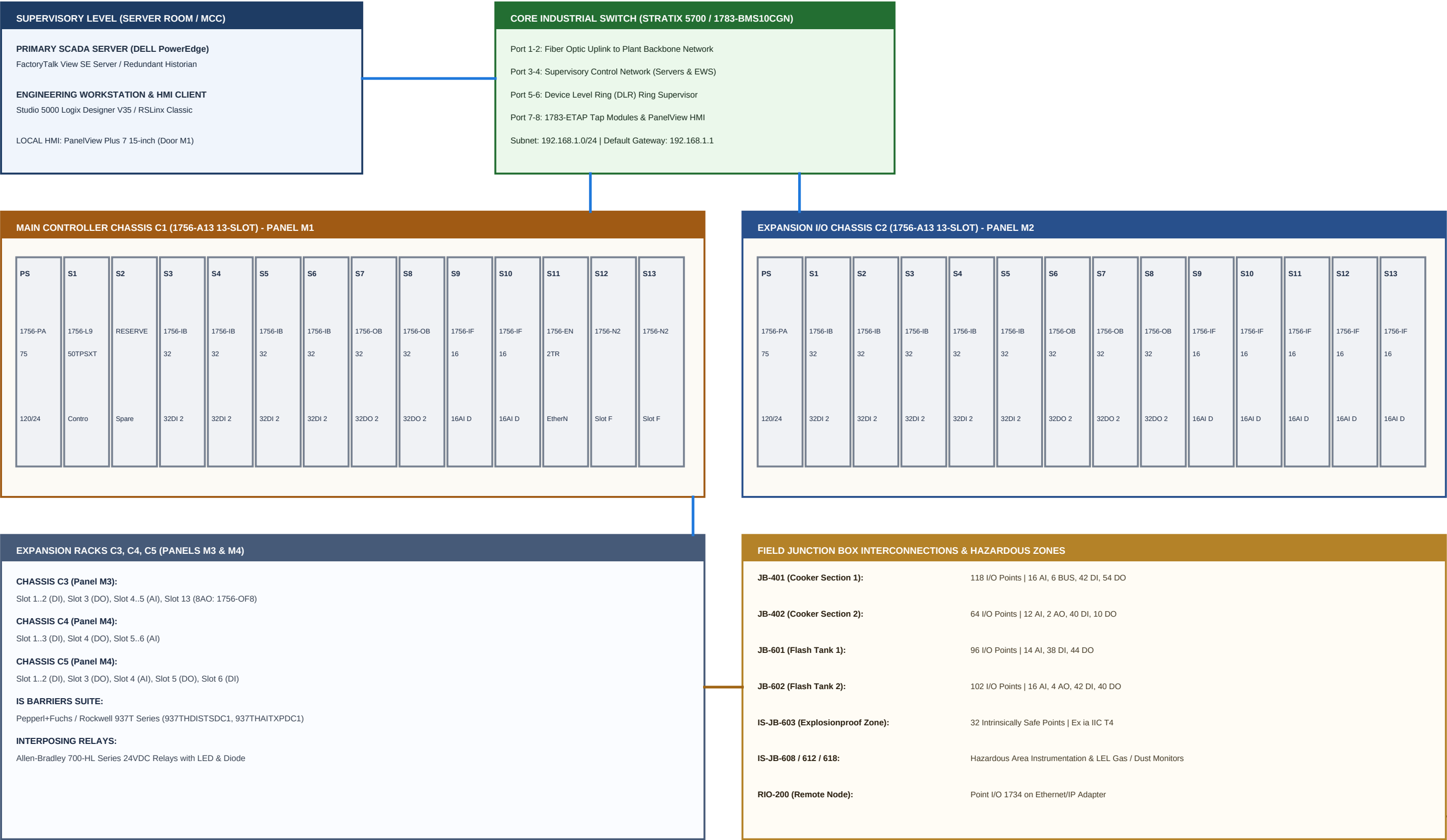
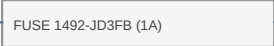
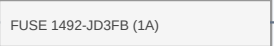
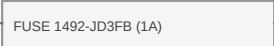
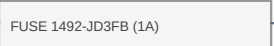
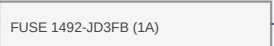
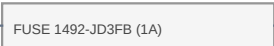
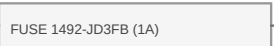
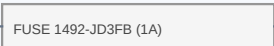
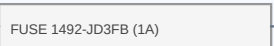
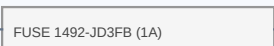
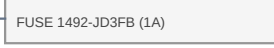
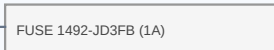
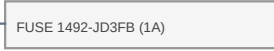
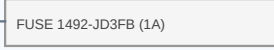
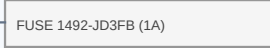
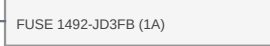
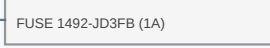
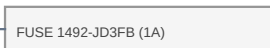
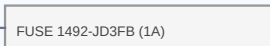
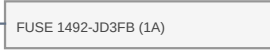


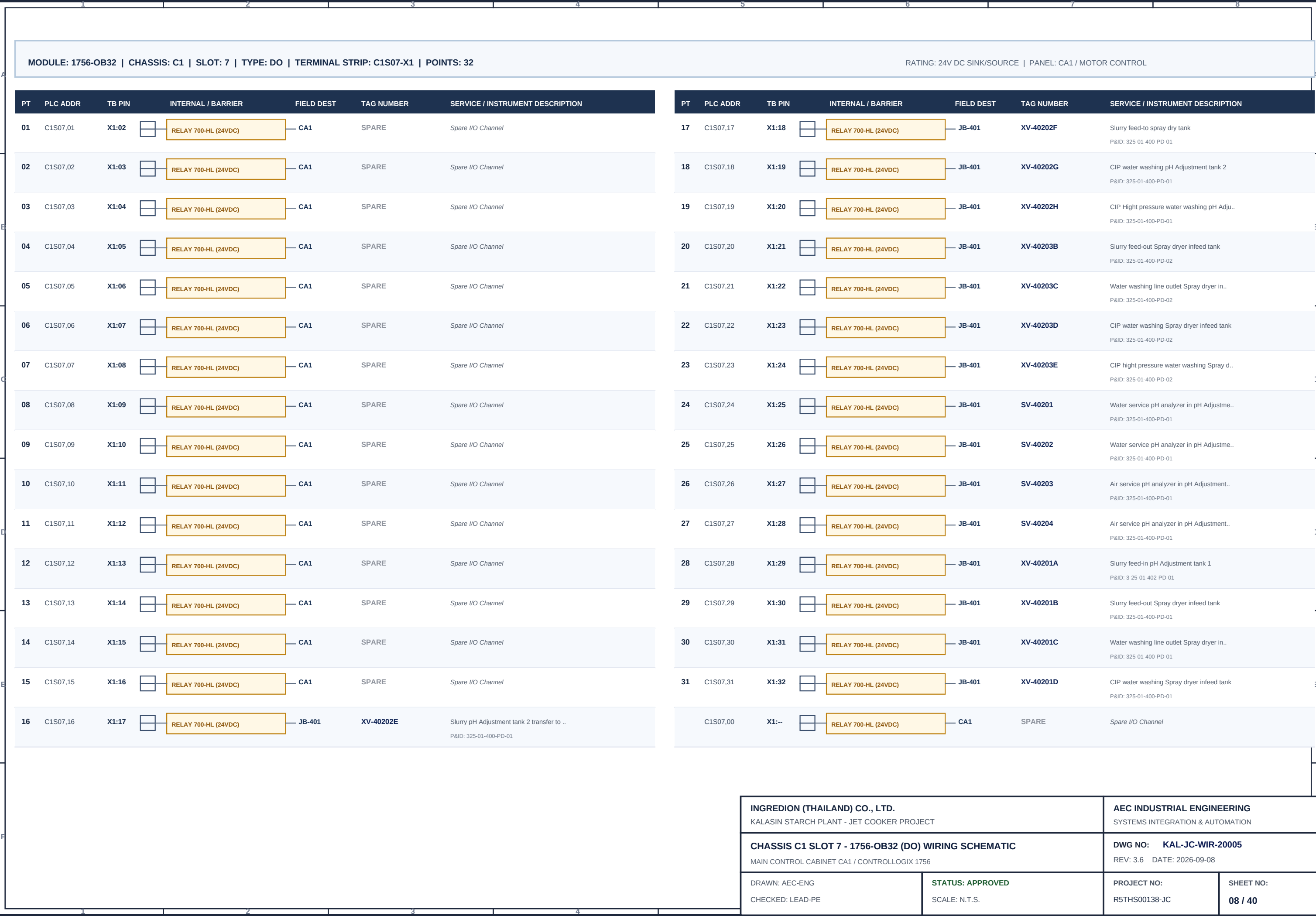
KALASIN JET COOKER AUTOMATION ARCHITECTURE - CONTROLLOGIX DLR & RIO

High-Availability Redundant Fiber Optic Device Level Ring (DLR) & Stratix Switch Network



MODULE: 1756-IB32 CHASSIS: C1 SLOT: 3 TYPE: DI TERMINAL STRIP: C1S03-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C1S03,01	X1:02		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	17	C1S03,17	X1:18		 FUSE 1492-JD3FB (1A)	JB-401	FSL-40202	Slurry low flow P&ID: 3-25-01-402-PD-01
02	C1S03,02	X1:03		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	18	C1S03,18	X1:19		 FUSE 1492-JD3FB (1A)	JB-401	FT-40201	Slurry flowrate & density check transfer.. P&ID: 325-01-400-PD-02
03	C1S03,03	X1:04		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	19	C1S03,19	X1:20		 FUSE 1492-JD3FB (1A)	JB-401	FT-40202	Feed Slurry Flow Transmitter for Pipe (P.. P&ID: 325-01-400-PD-02
04	C1S03,04	X1:05		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	20	C1S03,20	X1:21		 FUSE 1492-JD3FB (1A)	JB-401	HV-40201	Slurry Drain line P&ID: 325-01-400-PD-01
05	C1S03,05	X1:06		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	21	C1S03,21	X1:22		 FUSE 1492-JD3FB (1A)	JB-401	HV-40202	Slurry Drain line P&ID: 325-01-400-PD-01
06	C1S03,06	X1:07		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	22	C1S03,22	X1:23		 FUSE 1492-JD3FB (1A)	JB-401	HV-40203	Slurry Drain tank TS-40201 P&ID: 325-01-400-PD-01
07	C1S03,07	X1:08		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	23	C1S03,23	X1:24		 FUSE 1492-JD3FB (1A)	JB-401	HV-40204	Slurry Drain tank TS-40202 P&ID: 325-01-400-PD-01
08	C1S03,08	X1:09		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	24	C1S03,24	X1:25		 FUSE 1492-JD3FB (1A)	JB-401	HV-40205	Slurry Drain line P&ID: 325-01-400-PD-01
09	C1S03,09	X1:10		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	25	C1S03,25	X1:26		 FUSE 1492-JD3FB (1A)	JB-401	HV-40206	Slurry Drain tank TS-40203 P&ID: 325-01-400-PD-01
10	C1S03,10	X1:11		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	26	C1S03,26	X1:27		 FUSE 1492-JD3FB (1A)	JB-401	XV-40201A	Slurry feed-in pH Adjustment tank 1 P&ID: 3-25-01-402-PD-01
11	C1S03,11	X1:12		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	27	C1S03,27	X1:28		 FUSE 1492-JD3FB (1A)	JB-401	XV-40201A	Slurry feed-in pH Adjustment tank 1 P&ID: 3-25-01-402-PD-01
12	C1S03,12	X1:13		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	28	C1S03,28	X1:29		 FUSE 1492-JD3FB (1A)	JB-401	XV-40201B	Slurry feed-out Spray dryer infeed tank P&ID: 325-01-400-PD-01
13	C1S03,13	X1:14		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	29	C1S03,29	X1:30		 FUSE 1492-JD3FB (1A)	JB-401	XV-40201B	Slurry feed-out Spray dryer infeed tank P&ID: 325-01-400-PD-01
14	C1S03,14	X1:15		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	30	C1S03,30	X1:31		 FUSE 1492-JD3FB (1A)	JB-401	XV-40201C	Water washing line outlet Spray dryer in.. P&ID: 325-01-400-PD-01
15	C1S03,15	X1:16		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel	31	C1S03,31	X1:32		 FUSE 1492-JD3FB (1A)	JB-401	XV-40201C	Water washing line outlet Spray dryer in.. P&ID: 325-01-400-PD-01
16	C1S03,16	X1:17		 FUSE 1492-JD3FB (1A)	JB-401	FSL-40201	Slurry low flow P&ID: 3-25-01-402-PD-01		C1S03,00	X1:--		 FUSE 1492-JD3FB (1A)	CA1	SPARE	Spare I/O Channel
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT								AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION							
CHASSIS C1 SLOT 3 - 1756-IB32 (DI) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756								DWG NO: KAL-JC-WIR-20001 REV: 3.6 DATE: 2026-09-08							
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.				PROJECT NO: R5THS00138-JC		SHEET NO: 04 / 40					

MODULE: 1756-IB32 CHASSIS: C1 SLOT: 5 TYPE: DI TERMINAL STRIP: C1S05-X1 POINTS: 32																RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL																																															
PT								PLC ADDR								TB PIN								INTERNAL / BARRIER								FIELD DEST								TAG NUMBER								SERVICE / INSTRUMENT DESCRIPTION															
01								C1S05,01								X1:02																 FUSE 1492-JD3FB (1A)								JB-401								XV-40203E								CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-02							
02								C1S05,02								X1:03																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
03								C1S05,03								X1:04																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
04								C1S05,04								X1:05																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
05								C1S05,05								X1:06																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
06								C1S05,06								X1:07																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
07								C1S05,07								X1:08																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
08								C1S05,08								X1:09																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
09								C1S05,09								X1:10																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
10								C1S05,10								X1:11																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
11								C1S05,11								X1:12																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
12								C1S05,12								X1:13																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
13								C1S05,13								X1:14																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
14								C1S05,14								X1:15																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
15								C1S05,15								X1:16																 FUSE 1492-JD3FB (1A)								JB-401								SPARE								Spare I/O Channel							
16								C1S05,16								X1:17																 FUSE 1492-JD3FB (1A)								JB-601								ES-60202								E-Stop for PUPD-11 (Positive Displaceme.. P&ID: 325-01-600-PD-01							
17								C1S05,17								X1:18																 FUSE 1492-JD3FB (1A)								JB-601								FS-60205								Cooling Water Return Flow Switch for Pos.. P&ID: 325-01-600-PD-01							
18								C1S05,18								X1:19																 FUSE 1492-JD3FB (1A)								JB-601								LSL-60201								Level low switch P&ID: 325-01-600-PD-11							
19								C1S05,19								X1:20																 FUSE 1492-JD3FB (1A)								JB-601								SV-60201								PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08							
20								C1S05,20								X1:21																 FUSE 1492-JD3FB (1A)								JB-601								SV-60201								PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08							
21								C1S05,21								X1:22																 FUSE 1492-JD3FB (1A)								JB-601								SV-60202								PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08							
22								C1S05,22								X1:23																 FUSE 1492-JD3FB (1A)								JB-601								SV-60202								PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08							
23								C1S05,23								X1:24																 FUSE 1492-JD3FB (1A)								JB-601								SV-60203								PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08							
24								C1S05,24								X1:25																 FUSE 1492-JD3FB (1A)								JB-601								SV-60203								PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08							
25								C1S05,25								X1:26																 FUSE 1492-JD3FB (1A)								JB-601								VS-60205								Vibration Gear Box Vibration Switch for ... P&ID: 325-01-600-PD-01							
26								C1S05,26								X1:27																 FUSE 1492-JD3FB (1A)								JB-601								SPARE								Spare I/O Channel							
27								C1S05,27								X1:28																 FUSE 1492-JD3FB (1A)								JB-601								SPARE								Spare I/O Channel							
28								C1S05,28								X1:29																 FUSE 1492-JD3FB (1A)								JB-601								SPARE								Spare I/O Channel							
29								C1S05,29								X1:30																 FUSE 1492-JD3FB (1A)								JB-601								SPARE								Spare I/O Channel							
30								C1S05,30								X1:31																 FUSE 1492-JD3FB (1A)								JB-601								SPARE								Spare I/O Channel							
31								C1S05,31								X1:32																 FUSE 1492-JD3FB (1A)								JB-601								SPARE								Spare I/O Channel							
								C1S05,00								X1:--																 FUSE 1492-JD3FB (1A)								JB-401								XV-40203E								CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-02							



MODULE: 1756-OB32 CHASSIS: C1 SLOT: 8 TYPE: DO TERMINAL STRIP: C1S08-X1 POINTS: 32								RATING: 24V DC SINK/SOURCE PANEL: CA1 / MOTOR CONTROL							
PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION	PT	PLC ADDR	TB PIN		INTERNAL / BARRIER	FIELD DEST	TAG NUMBER	SERVICE / INSTRUMENT DESCRIPTION
01	C1S08,01	X1:02		RELAY 700-HL (24VDC)	JB-401	XV-40201F	Slurry feed-to spray dry tank P&ID: 325-01-400-PD-01	17	C1S08,17	X1:18		RELAY 700-HL (24VDC)	JB-601	SV-60202	PA On/Off Valve for Cyclone-22 Rotary Va.. P&ID: 325-01-600-PD-08
02	C1S08,02	X1:03		RELAY 700-HL (24VDC)	JB-401	XV-40201G	CIP water washing pH Adjustment tank 1 P&ID: 325-01-400-PD-01	18	C1S08,18	X1:19		RELAY 700-HL (24VDC)	JB-601	SV-60230	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-04
03	C1S08,03	X1:04		RELAY 700-HL (24VDC)	JB-401	XV-40201H	CIP Hight pressure water washing pH Adju.. P&ID: 325-01-400-PD-01	19	C1S08,19	X1:20		RELAY 700-HL (24VDC)	JB-601	SV-60203	PA On/Off Valve for Bag House Rotary Val.. P&ID: 325-01-600-PD-08
04	C1S08,04	X1:05		RELAY 700-HL (24VDC)	JB-401	XV-40202A	Slurry feed-in pH Adjustment tank 2 P&ID: 325-01-400-PD-01	20	C1S08,20	X1:21		RELAY 700-HL (24VDC)	JB-601	SV-60240A	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-05
05	C1S08,05	X1:06		RELAY 700-HL (24VDC)	JB-401	XV-40202B	Slurry feed-out pH Adjustment tank 2 P&ID: 325-01-400-PD-01	21	C1S08,21	X1:22		RELAY 700-HL (24VDC)	JB-601	SV-60231	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-04
06	C1S08,06	X1:07		RELAY 700-HL (24VDC)	JB-401	XV-40202C	Water washing line outlet pH Adjustment .. P&ID: 325-01-400-PD-01	22	C1S08,22	X1:23		RELAY 700-HL (24VDC)	JB-601	SV-60240B	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-05
07	C1S08,07	X1:08		RELAY 700-HL (24VDC)	JB-401	XV-40202D	Slurry circulation pH Adjustment tank 2 P&ID: 325-01-400-PD-01	23	C1S08,23	X1:24		RELAY 700-HL (24VDC)	JB-601	SV-60240C	PA On/Off Valve for Cyclone Air Hammer 2.. P&ID: 325-01-600-PD-05
08	C1S08,08	X1:09		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	24	C1S08,24	X1:25		RELAY 700-HL (24VDC)	JB-601	XV-60240D	Instrument XV-60240D P&ID: 325-01-600-PD-05
09	C1S08,09	X1:10		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	25	C1S08,25	X1:26		RELAY 700-HL (24VDC)	JB-601	XV-60251	Liquid On/Off Valve for Drain Pipe (PIPE) P&ID: 325-01-600-PD-01
10	C1S08,10	X1:11		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	26	C1S08,26	X1:27		RELAY 700-HL (24VDC)	JB-601	XV-60202	High Pressure Water On/Off Valve for Dra.. P&ID: 325-01-600-PD-11
11	C1S08,11	X1:12		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	27	C1S08,27	X1:28		RELAY 700-HL (24VDC)	JB-601	XV-60205A	Cooling Water On/Off Valve for Cooling W.. P&ID: 325-01-600-PD-01
12	C1S08,12	X1:13		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	28	C1S08,28	X1:29		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
13	C1S08,13	X1:14		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	29	C1S08,29	X1:30		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
14	C1S08,14	X1:15		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	30	C1S08,30	X1:31		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
15	C1S08,15	X1:16		RELAY 700-HL (24VDC)	JB-401	SPARE	Spare I/O Channel	31	C1S08,31	X1:32		RELAY 700-HL (24VDC)	JB-601	SPARE	Spare I/O Channel
16	C1S08,16	X1:17		RELAY 700-HL (24VDC)	JB-601	SV-60201	PA On/Off Valve for Cyclone-21 Rotary Va.. P&ID: 325-01-600-PD-08		C1S08,00	X1:--		RELAY 700-HL (24VDC)	JB-401	XV-40201E	CIP hight pressure water washing Spray d.. P&ID: 325-01-400-PD-01
INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT								AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION							
CHASSIS C1 SLOT 8 - 1756-OB32 (DO) WIRING SCHEMATIC MAIN CONTROL CABINET CA1 / CONTROLLOGIX 1756								DWG NO: KAL-JC-WIR-20006 REV: 3.6 DATE: 2026-09-08							
DRAWN: AEC-ENG CHECKED: LEAD-PE				STATUS: APPROVED SCALE: N.T.S.				PROJECT NO: R5THS00138-JC		SHEET NO: 09 / 40					

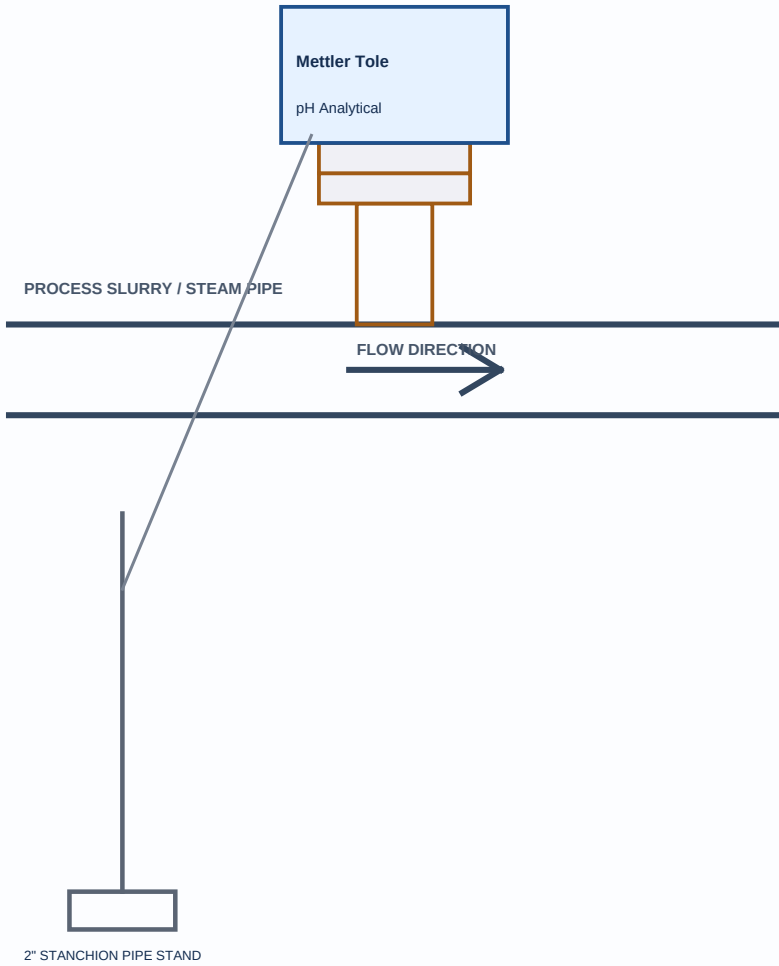
MODULE: 1756-IF16 CHASSIS: C1 SLOT: 9 TYPE: AI TERMINAL STRIP: C1S09-X1 POINTS: 16																RATING: 4-20mA CURRENT LOOP (ISOLATED) PANEL: CA1 / MOTOR CONTROL															

TEMPLATE TYP-03: PH ANALYTICAL LOOP

TYPICAL TAGS: AIT-40101 / AIT-40201

SERVICE: Cooker Slurry & Flash Tank pH Neutralization | PLC I/O: 1756-IF16 (AI pH) & 1756-IB32 (DI Alarms)

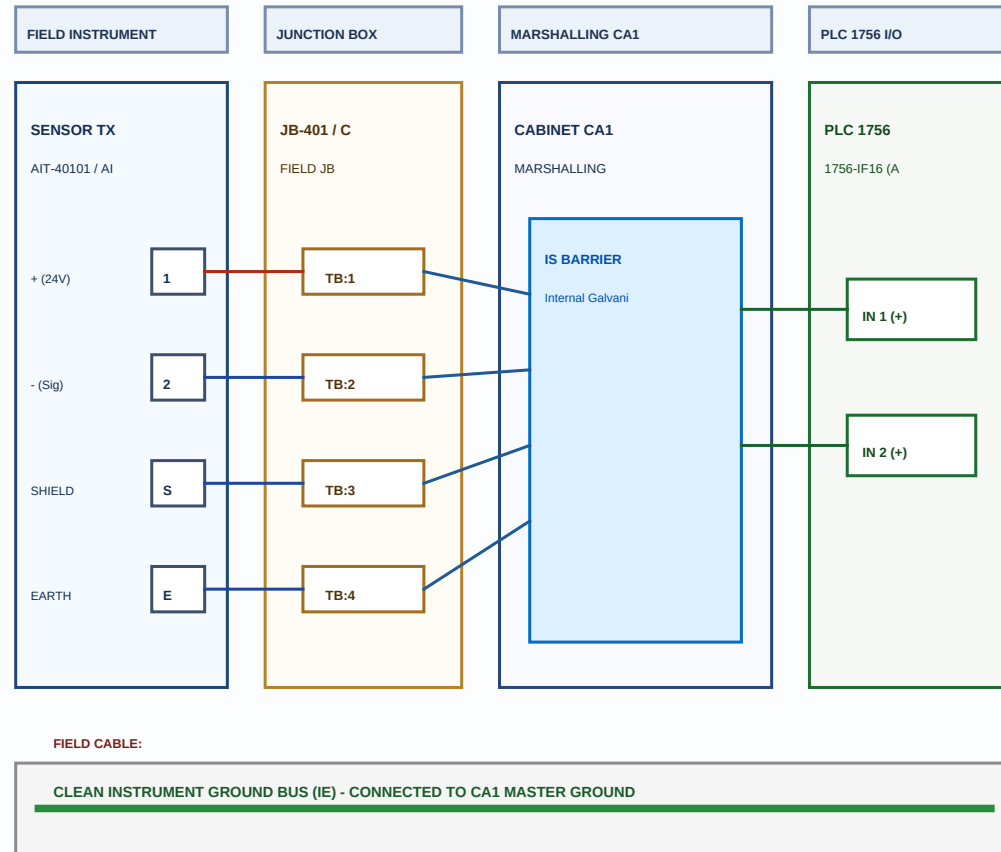
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN25 Weld-in Socket (Incline..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP65 Polycarbonate..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6/T4..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	AI: 0-14 pH (4-20mA HART) / AI: Temp
Power Supply:	24V DC / 10W (Optional 100-240V AC)
Field Cable Type:	VP6 High-Z Cable to Tx + 2Px1.5mm² Shi...
Destination JB:	JB-401 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation (4-Wire Tx)
PLC Module:	1756-IF16 (AI pH) & 1756-IB32 (DI Alar...

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: METTLER TOLEDO

MODEL: M300G2 4-Wire Transmitter + InPro3250i + InTrac 777P

SERVICE: Cooker Slurry & Flash Tank pH Neutralization

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	pH Multi-Parameter Transmitter Mettler Toledo M300 4-Wire		1 EA
2	Intelligent pH Sensor InPro325 Pre-pressurized Liquid Gel		1 EA
3	Retractable Housing InTrac 777 SS316L / Hastelloy C w/ Flange		1 EA
4	Weld-in Socket 40mm / DN25 (15 SS316L Sloped Socket		1 EA
5	2" Pipe Mounting Kit for M300G SS304 Bracket & U-Bolts		1 SET

ENGINEERING INSTALLATION NOTES

1. Weld socket must be installed inclined at min. 15° above horizontal to prevent air bubble entrapment.
2. Use retractable housing safety interlock valve before extracting probe during process operation.
3. Ground transmitter chassis directly to plant PE bar with min. 4mm² copper grounding strap.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

pH ANALYTICAL MEASUREMENT LOOP HOOK-UP & LOOP DIAGRAM

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED

SCALE: N.T.S.

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

DWG NO: **KAL-JC-TYP-003**
REV: 3.6 DATE: 2026-09-08

PROJECT NO:
R5THS00138-JC

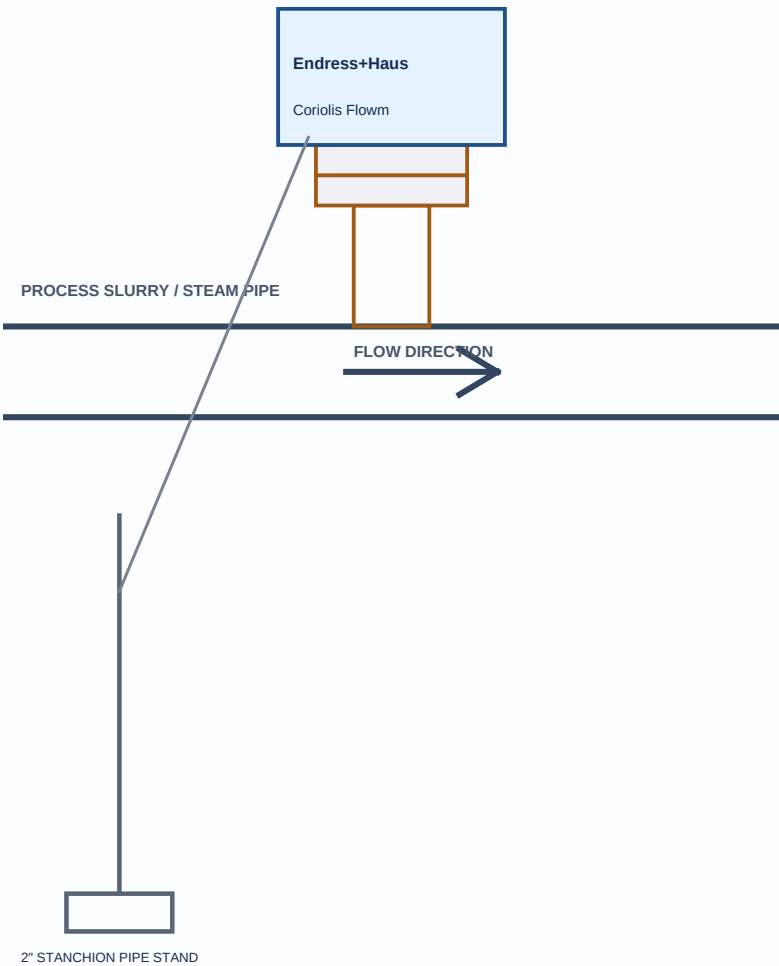
SHEET NO:
05 / 25

TEMPLATE TYP-06: CORIOLIS FLOWMETER

TYPICAL TAGS: FT-40201 Series

SERVICE: Starch Slurry High-Accuracy Mass Flow & Density | PLC I/O: 1756-IF16 (Ch. 1 Mass Flow / Ch. 2 Density)

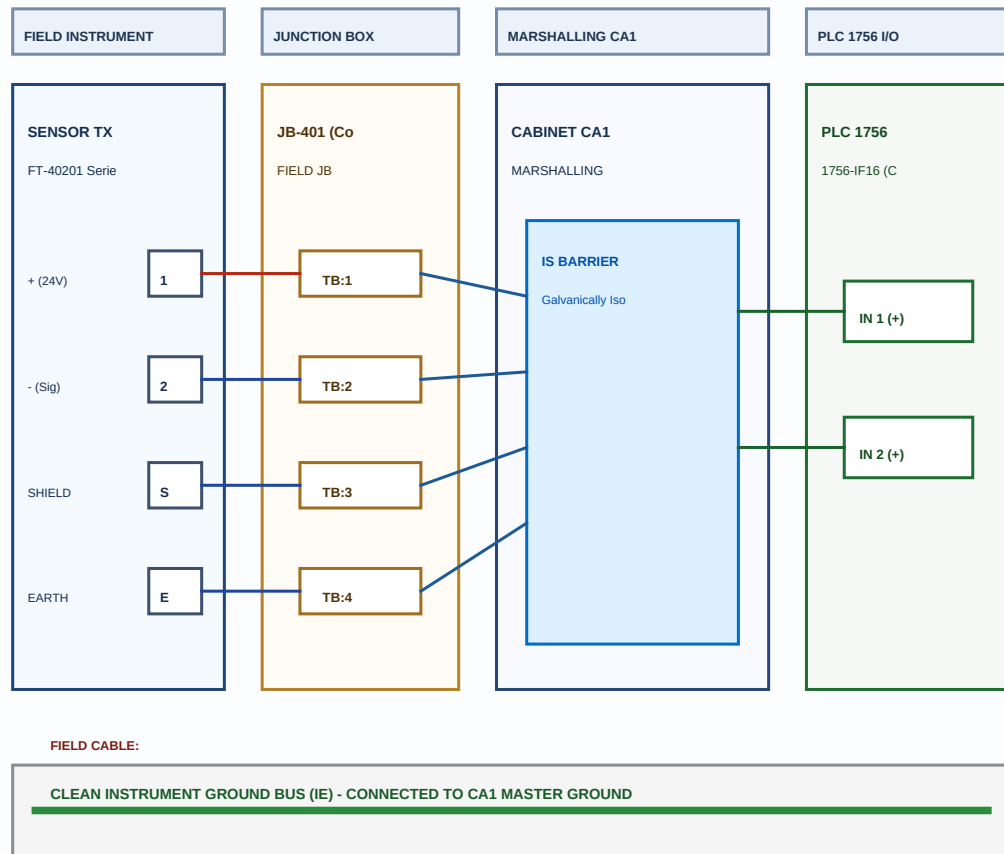
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50 (2") ASME B16.5 Cl.150 ..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP67 Die-cast Aluminum Trans..
Hazardous Class:	ATEX II 2G Ex db eb ia IIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Mass (kg/h) & Density (g/c..
Power Supply:	24V DC / 10W (Transmitter Electronics)
Field Cable Type:	2Px1.5mm² Shielded Overall (Belden 877..
Destination JB:	JB-401 (Coker Section 1)
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Galvanically Isolated Active 4-20mA Ou..
PLC Module:	1756-IF16 (Ch. 1 Mass Flow / Ch. 2 Den..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promass E 300 (8E3B50 / Dual Tube SS316L)

SERVICE: Starch Slurry High-Accuracy Mass Flow & Density

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promass E 300 Coriolis Flowmet E+H DN50 Cl.150 SS316L Dual Tube		1 EA
2	Companion Counter-Flanges DN50 ASTM A182 F316 / RF Flanges		2 EA
3	PTFE / Spiral Wound Gaskets 316SS with PTFE Filler		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		2 EA
5	Pipe Support Clamps & Structur Heavy Duty Rigidity Anchors		2 SET

ENGINEERING INSTALLATION NOTES

- Sensor tubes must be installed pointing downwards (flag-down) for liquid slurry to prevent gas collection.
- Support pipeline rigidly on both upstream and downstream sides within 2 pipe diameters of meter flanges.
- Do not clamp or attach pipe supports directly to the Coriolis sensor housing.

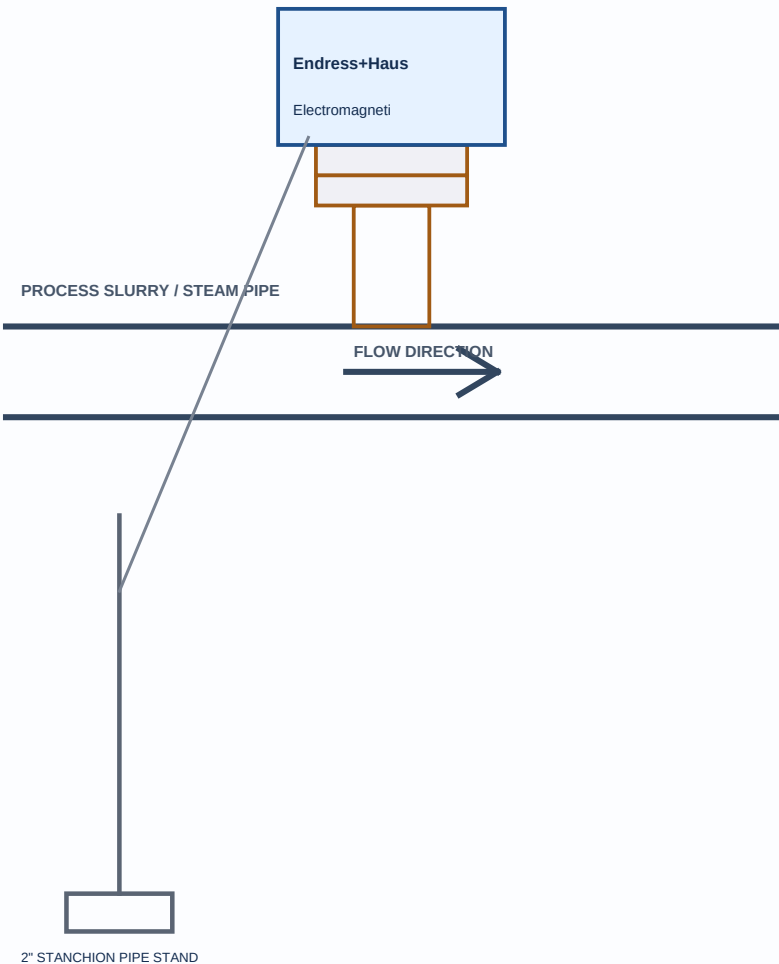
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONCORIOLIS MASS FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-006
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
08 / 25

TEMPLATE TYP-07: ELECTROMAGNETIC FLOWMETER

TYPICAL TAGS: FT-40202 / FT-60201

SERVICE: Slurry & Process Water Volumetric Flow Measurement | PLC I/O: 1756-IF16 (4-20mA Flow) & 1756-IB32 (Pulse Counter)

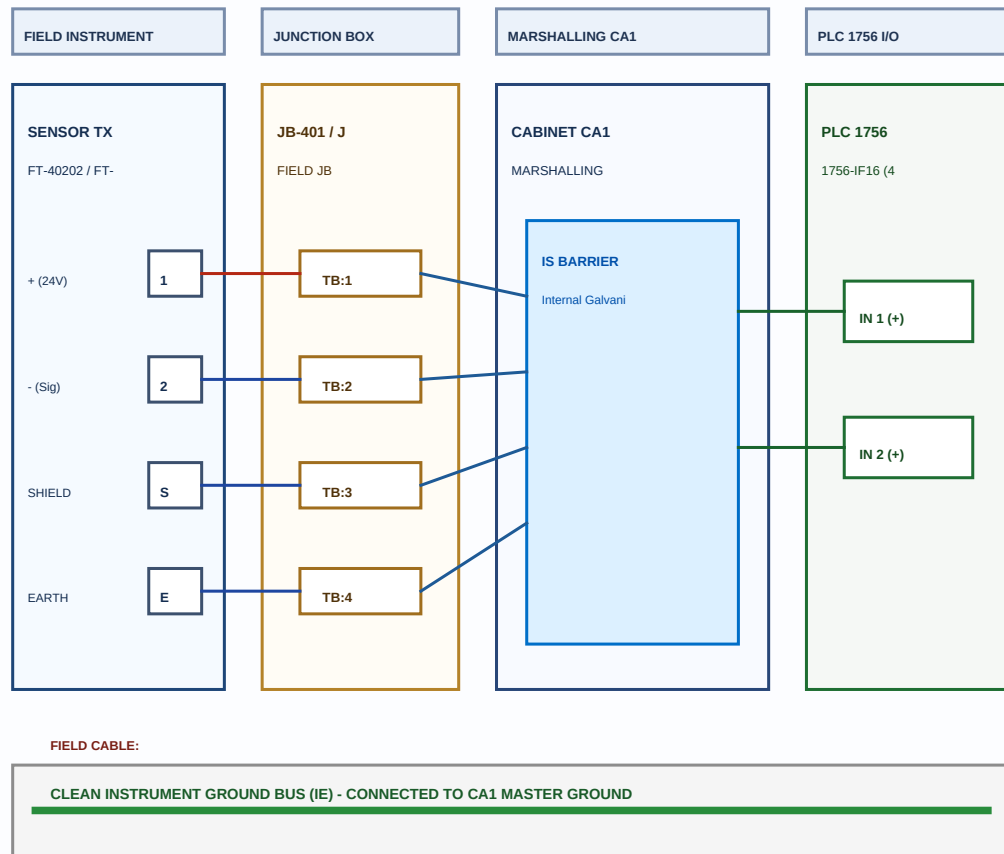
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50/80/100 ASME CL150 RF F..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Tran..
Hazardous Class:	ATEX II 2G Ex eb Ia IIC T6....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-150 m³/h) + Scaled Pulses
Power Supply:	24V DC / 8W (4-Wire Electronics)
Field Cable Type:	1Px1.5mm² (Analog) + 1Px1.5mm² (Pulse/..
Destination JB:	JB-401 / JB-601
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation
PLC Module:	1756-IF16 (4-20mA Flow) & 1756-IB32 (P..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promag P 300 (5P3B50 / 80 / 1H PTFE Liner)

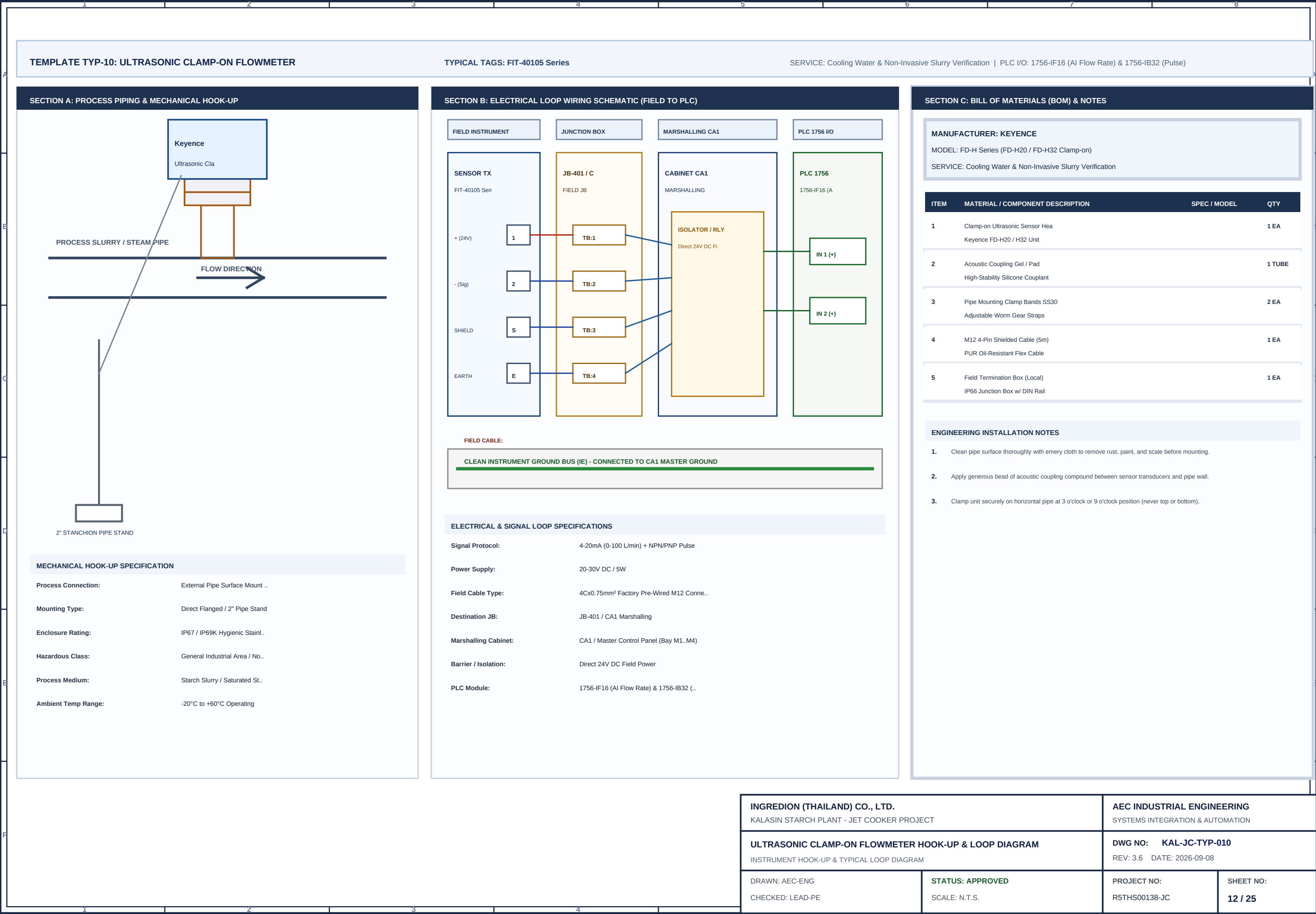
SERVICE: Slurry & Process Water Volumetric Flow Measurement

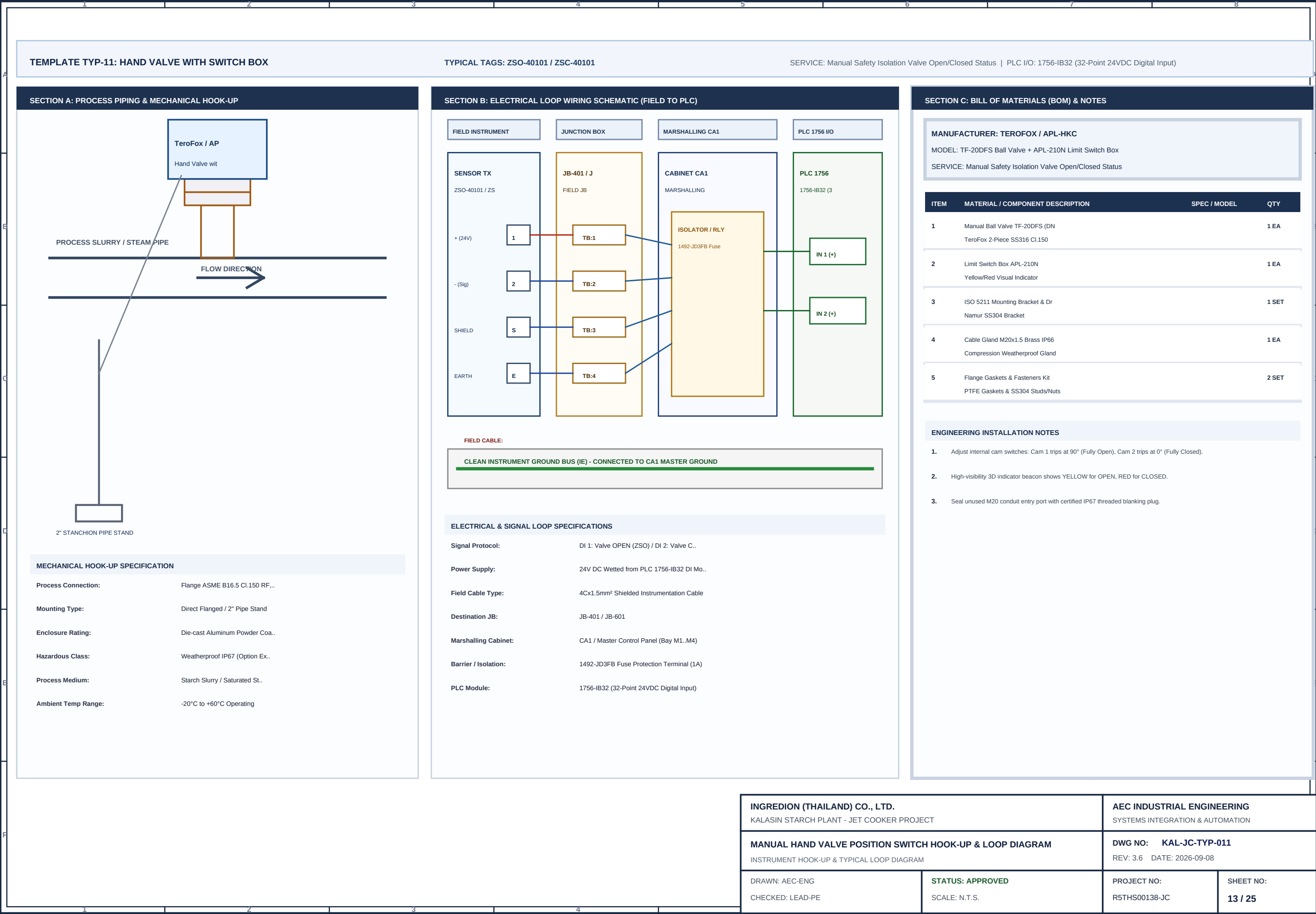
ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promag P 300 Magmeter Sensor & E+H PTFE Lined, Hastelloy C Elec		1 EA
2	Grounding Rings / Discs SS316L E+H Grounding Electrodes Kit		2 EA
3	PTFE Envelope Flange Gaskets Chemical Resistant Gaskets		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Glands		2 EA
5	Grounding Cables 6mm² Green/Ye Copper Earthing Strap with Lugs		2 EA

ENGINEERING INSTALLATION NOTES

- Straight piping requirements: min. 5D upstream and 2D downstream of flowmeter.
- Install grounding rings between meter flanges and pipe counter-flanges to ensure reference ground.
- Pipe must always remain 100% full of conductive liquid for accurate flow measurement.

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KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTROMAGNETIC FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-007
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
09 / 25



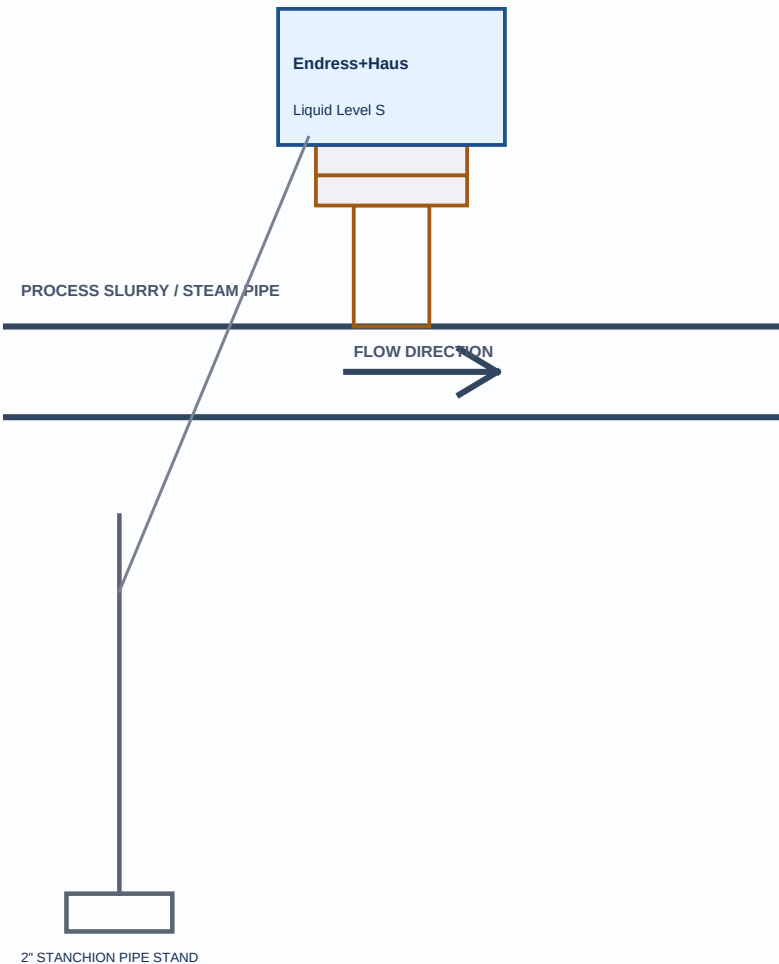


TEMPLATE TYP-12: LIQUID LEVEL SWITCH

TYPICAL TAGS: LSH-40101 / LSL-40101

SERVICE: High / Low Slurry Level Interlock & Pump Protection | PLC I/O: 1756-IB32 (24V DC Sinking/Sourcing Digital Input)

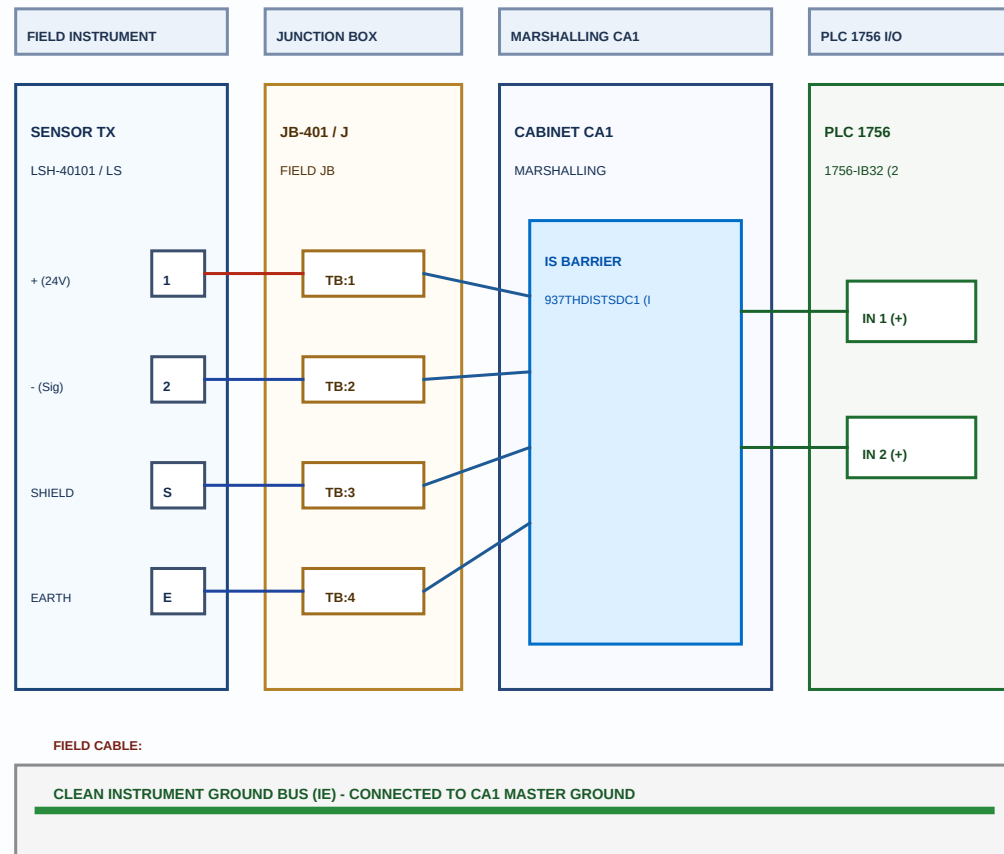
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	3/4" NPT Male Threaded / Tri..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Stainless ..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6.....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Level Alarm (Dry / Submerged ..
Power Supply:	24V DC (10-55V DC, PNP Mode)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THDISTSDC1 (IS Digital Input Barrier)
PLC Module:	1756-IB32 (24V DC Sinking/Sourcing Dig..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER
MODEL: Liquiphant FTL51B (Tuning Fork SS316L)
SERVICE: High / Low Slurry Level Interlock & Pump Protection

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Liquiphant FTL51B Level Switch E+H Tuning Fork SS316L (100mm)		1 EA
2	Welding Boss 3/4" NPT Half Cou SS316 Tank Welding Socket		1 EA
3	Cable Gland M20x1.5 Ex d/e Nickel-plated Brass Armored Glan		1 EA
4	Viton / EPDM Thread Seal Ring Process Sealing Gasket		1 EA
5	Grounding Strap 4mm² with Lugs Copper Flexible Earth Braid		1 EA

ENGINEERING INSTALLATION NOTES

- Install fork with tines aligned vertically to prevent slurry buildup on top of vibrating blades.
- Set fail-safe mode switch: MAX (Overfill Protection) or MIN (Dry Run Pump Protection).
- Keep fork at least 50mm away from tank walls, internal baffles, and agitator blades.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECT

AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATION

LIQUID LEVEL SWITCH (VIBRATING FORK) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAM

DWG NO: KAL-JC-TYP-012
REV: 3.6 DATE: 2026-09-08

DRAWN: AEC-ENG
CHECKED: LEAD-PE

STATUS: APPROVED
SCALE: N.T.S.

PROJECT NO:
R5THS00138-JC

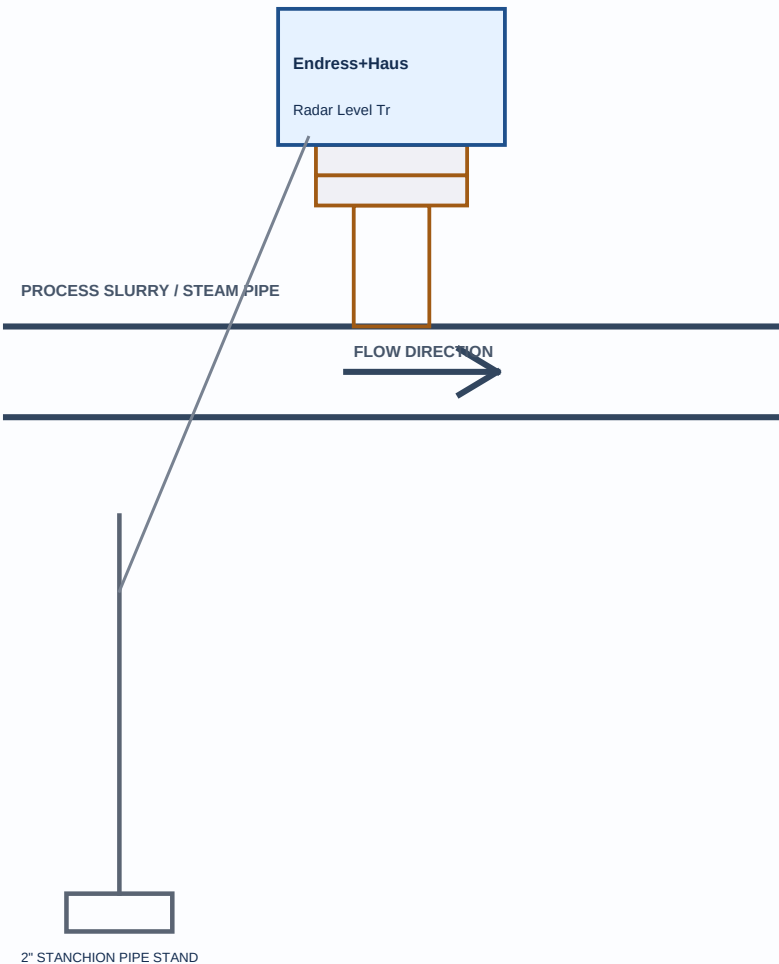
SHEET NO:
14 / 25

TEMPLATE TYP-14: RADAR LEVEL TRANSMITTER

TYPICAL TAGS: LT-40101 / LT-40201

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level | PLC I/O: 1756-IF16 (16-Point Differential AI)

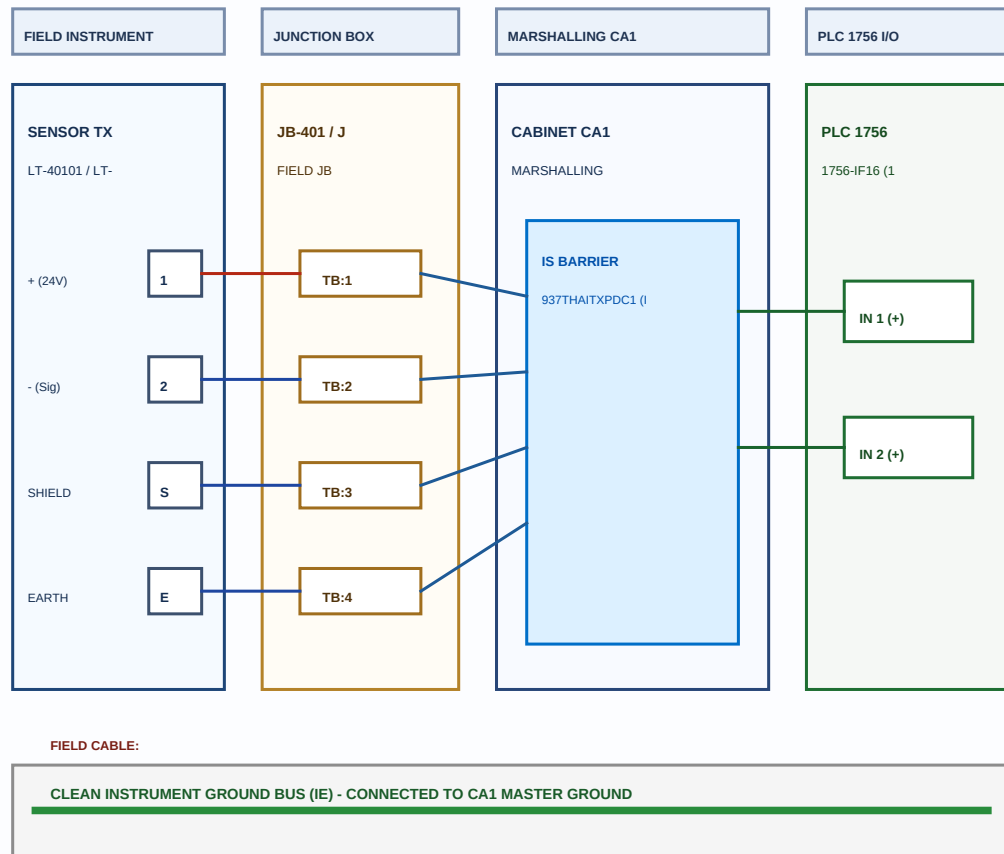
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80 (3") ASME Cl.150 RF Fla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Level (0-100% / 0-6.0m)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Pair Shielded (Belde..
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Micropilot FMR67B (80 GHz High-Frequency Radar)

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Micropilot FMR67B Radar Transm E+H 80GHz w/ Drip-off PTFE Anten		1 EA
2	Tank Nozzle Flange DN80 Cl.150 SS316 Tank Flange Neck		1 EA
3	PTFE Envelope Gasket DN80 Cl.1 Chemical Resistant Gasket		1 EA
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Gland		1 EA
5	Air Purge Connection Kit 1/4" Purge Connection for Condensate		1 SET

ENGINEERING INSTALLATION NOTES

- Mount nozzle height must not exceed antenna length so horn protrudes clear into vessel interior.
- Distance to vessel wall should be min. 1/6 of vessel diameter to prevent false sidewall echoes.
- Activate Heartbeat Technology and mapping curves to mask agitator blade echoes during commissioning.

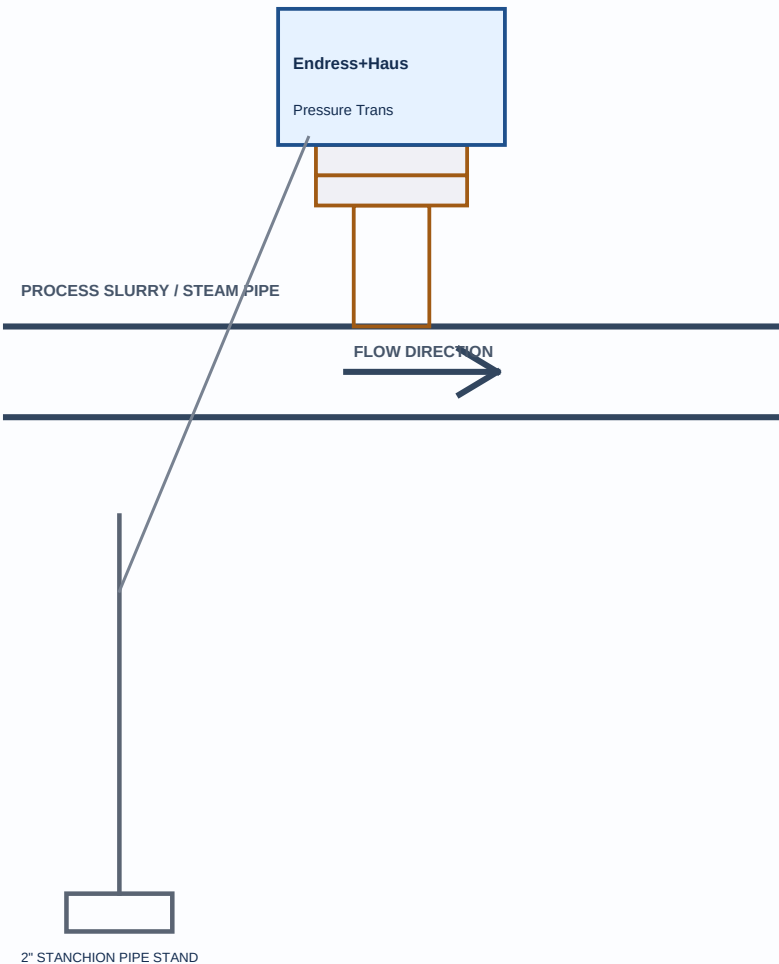
INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONRADAR LEVEL TRANSMITTER (80 GHz) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-014
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
16 / 25

TEMPLATE TYP-17: PRESSURE TRANSMITTER

TYPICAL TAGS: PT-40101 / PT-60202

SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank | PLC I/O: 1756-IF16 (16-Point Differential AI)

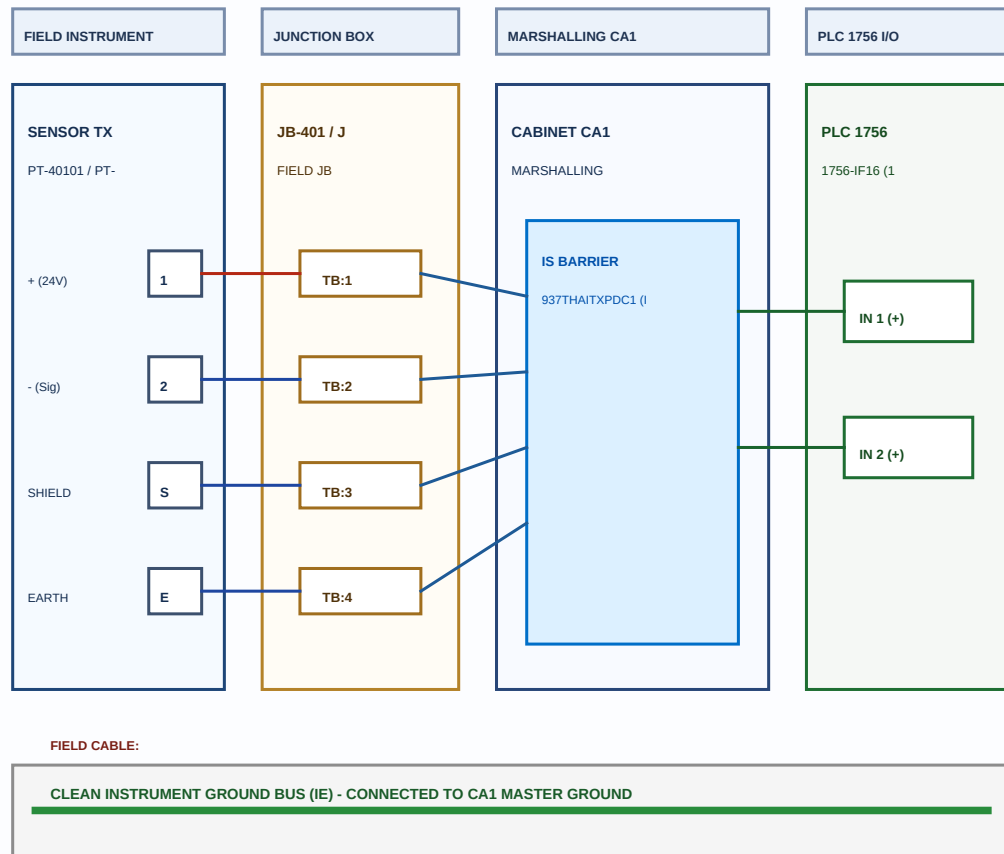
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Thread / 2" Tr..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-10 bar / 0-16 bar gauge)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (Belden 8761)
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Cerabar PMP51B / PMC51B (Metallic / Ceramic)

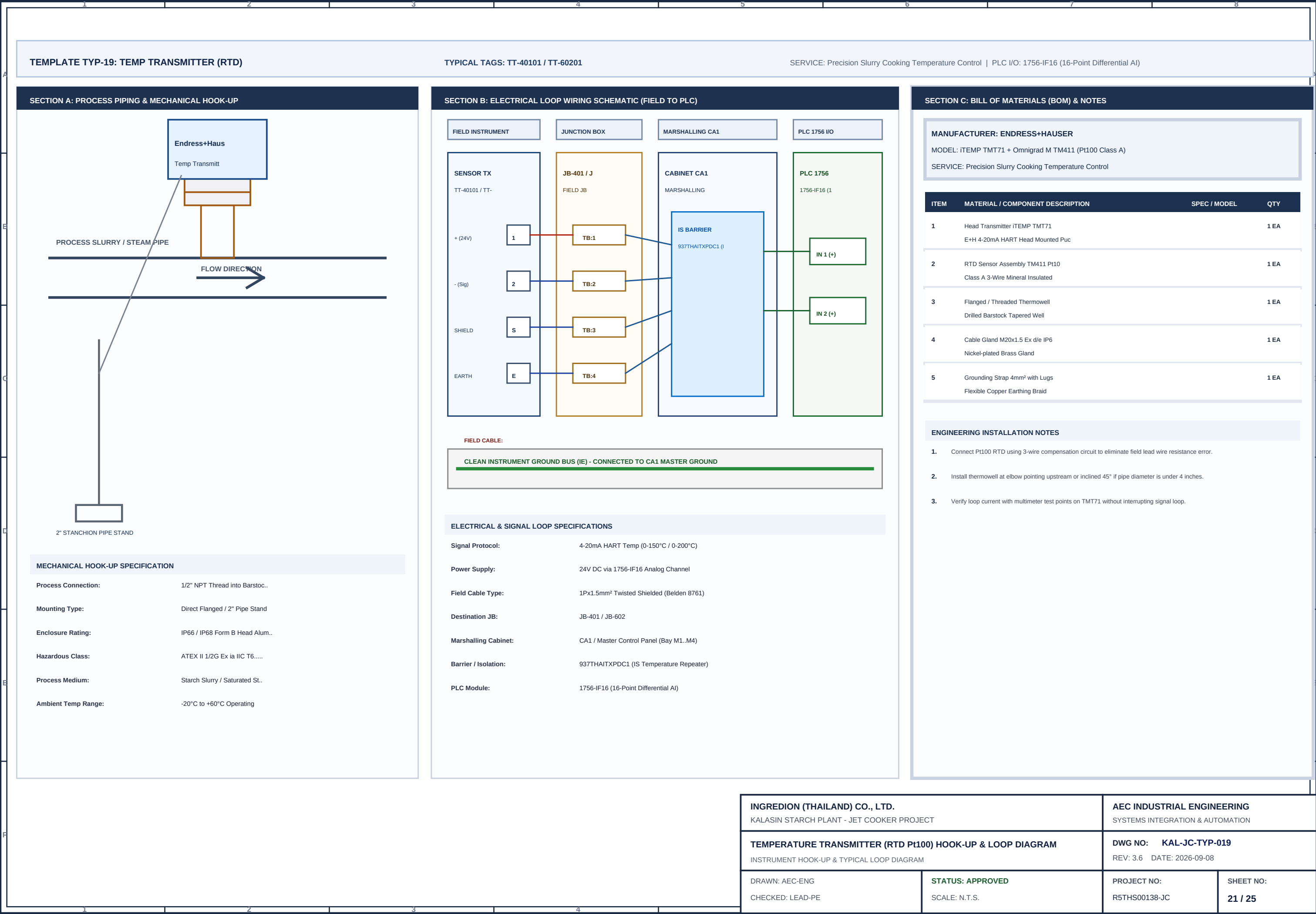
SERVICE: Continuous Pressure Monitoring for Cooker & Flash Tank

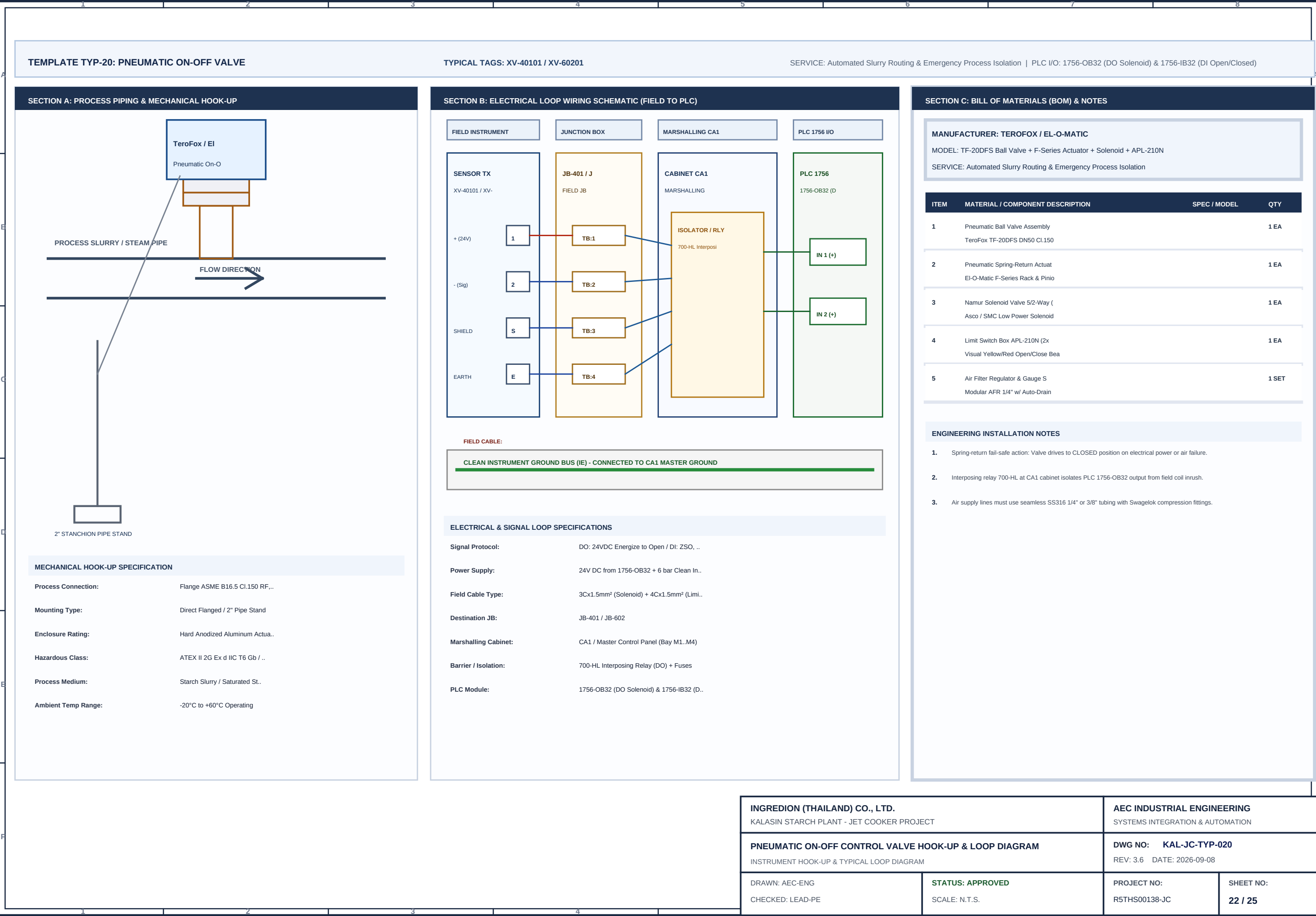
ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Cerabar PMP51B Pressure Transm E+H Piezoresistive SS316L Cell		1 EA
2	2-Valve Block & Bleed Manifold SS316 1/2" NPT Male x Female		1 EA
3	2" Pipe Mounting Bracket Kit SS304 Universal Bracket & U-Bolt		1 SET
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Armored Gland		1 EA
5	Siphon / Cooling Neck (Steam D SS316 Pigtail Loop 1/2" NPT		1 EA

ENGINEERING INSTALLATION NOTES

- Use flush diaphragm seal (PMC51B) for thick starch slurry to eliminate clogging in impulse ports.
- Ground transmitter body to local pipe ground; shield wire grounded only at CA1 cabinet earth bar.
- HART configuration: Set damping to 1.0s to filter process pump pulsations.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONPRESSURE TRANSMITTER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-017
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
19 / 25



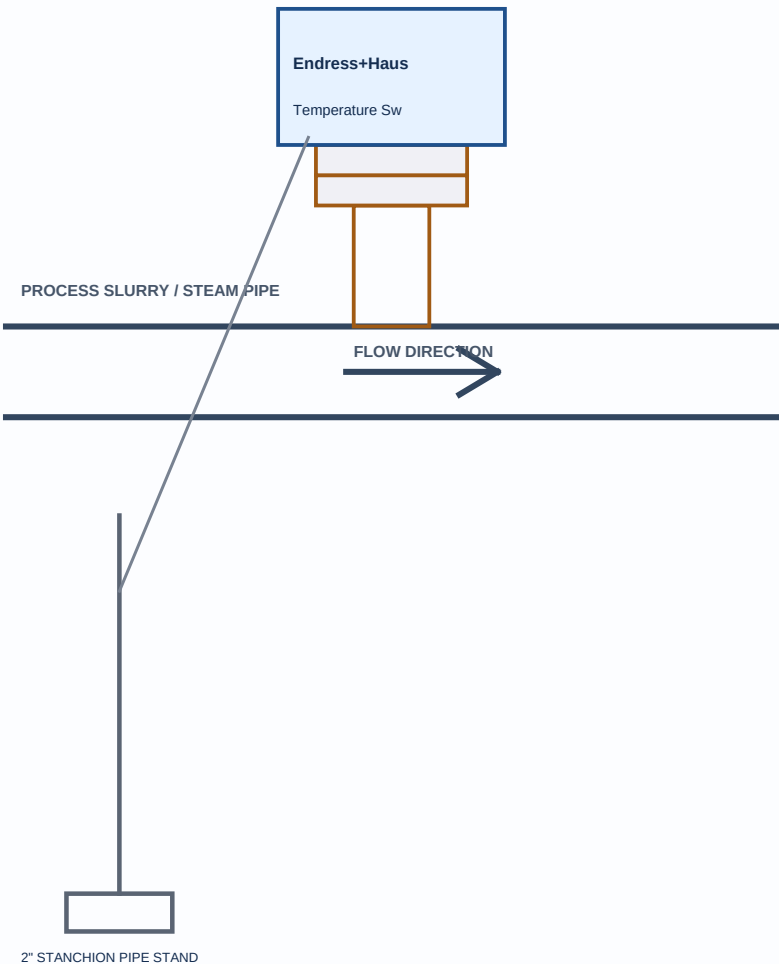


TEMPLATE TYP-22: TEMPERATURE SWITCH

TYPICAL TAGS: TSH-40101 / TSH-60201

SERVICE: High Temperature Safety Trip & Alarm | PLC I/O: 1756-IB32 (24V DC Digital Input)

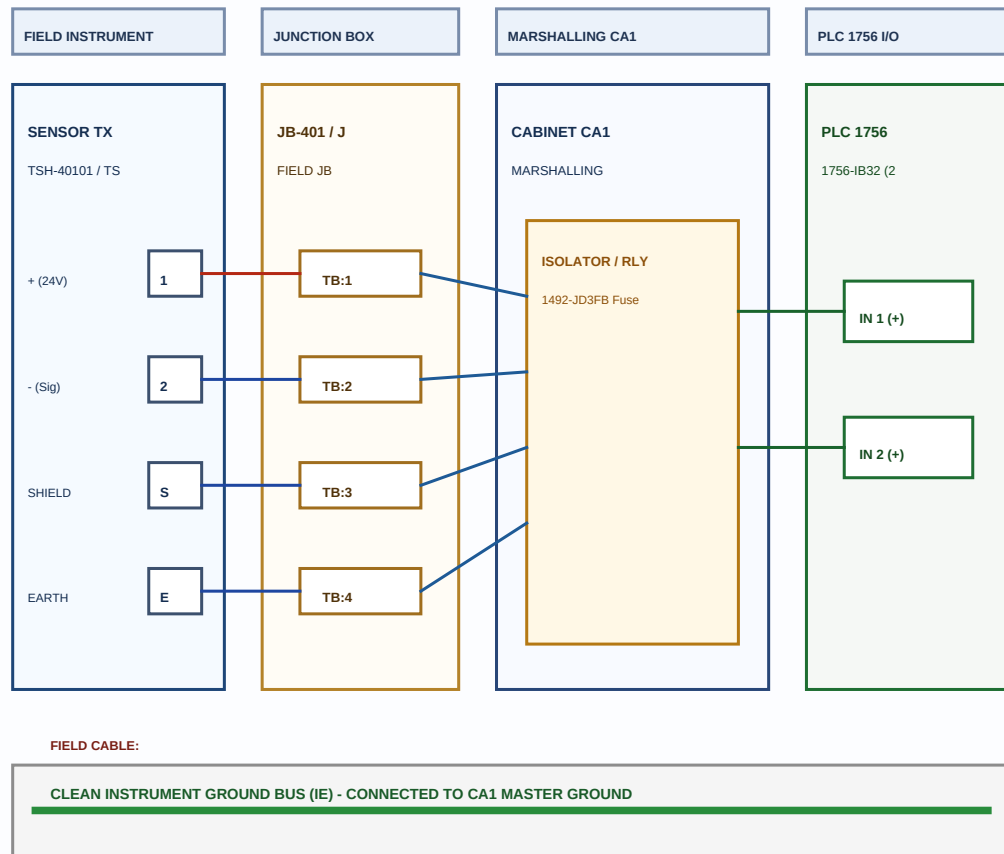
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Threaded / Hyg..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 / IP67 Stainless Steel ..
Hazardous Class:	General Industrial / Hygieni..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Temperature Alarm Trip
Power Supply:	24V DC (10-30V DC, 35mA)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Protection Terminal
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Thermophant TTR35 (Compact Pt100 Switch)

SERVICE: High Temperature Safety Trip & Alarm

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Thermophant TTR35 Temp Switch E+H Pt100 Class A, Digital Displ		1 EA
2	Barstock Thermowell SS316L (U= 1/2" NPT x 1/2" NPT Process Well		1 EA
3	M12 4-Pin Female Cable Assembl M12 Right Angle Molded Connector		1 EA
4	Thredolet / Boss 1/2" NPT SS3 Pipe Welding Fitting		1 EA
5	Grounding Clamp & Lug Copper Earth Wire		1 SET

ENGINEERING INSTALLATION NOTES

- Configure switch point via optical touch keys on front cover without opening enclosure.
- Normally closed (NC) contact output configuration guarantees fail-safe alarm on wire break.
- Thermowell allows switch replacement without draining the process line.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTRONIC TEMPERATURE SWITCH HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-022
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
24 / 25

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E

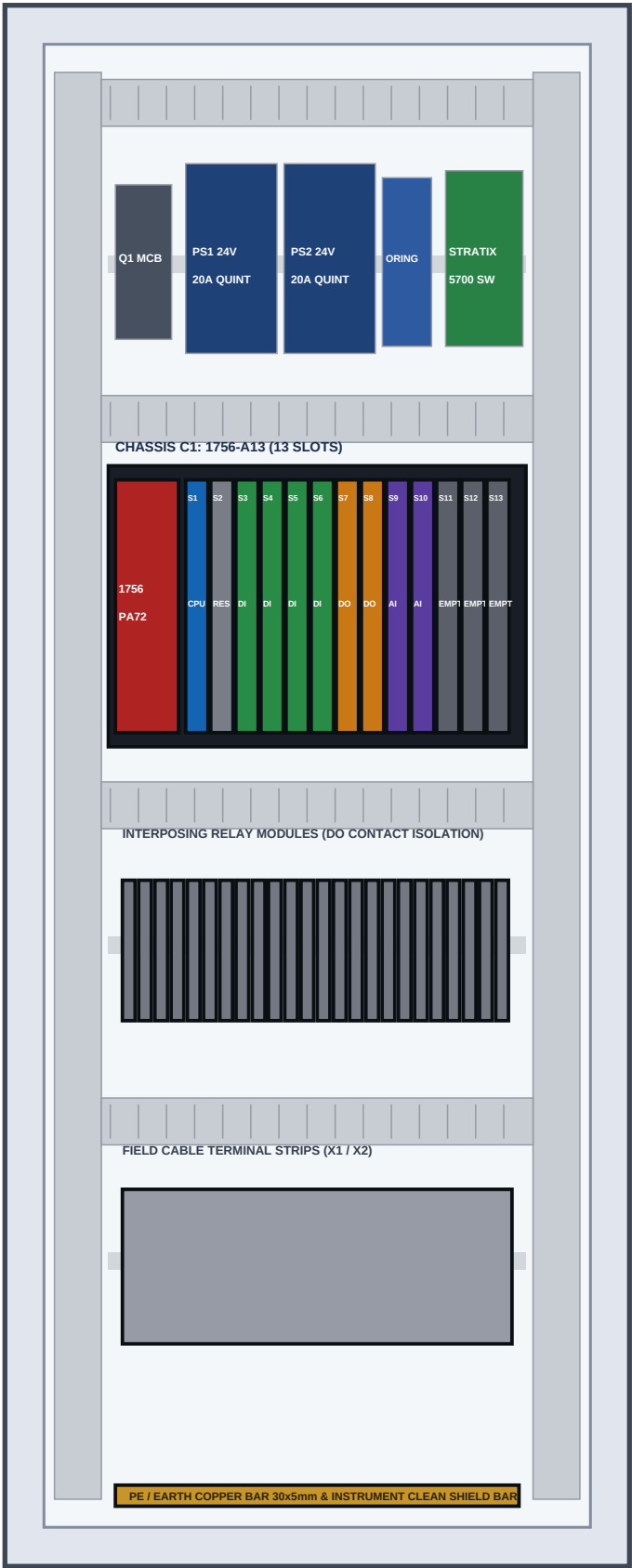
F



NOTES & DESIGN CRITERIA			
1. CHASSIS SYSTEM: Rockwell Automation Allen-Bradley ControlLogix 1756-A13 (13-slot racks).			
2. WIRING SEGREGATION: 50mm min. clearance between 230VAC, 24VDC, and analog signal paths.			
3. INTRINSIC SAFETY: Panels M3 & M4 feature dedicated blue IS terminal strips and 50mm physical air gap partition plates complying with IEC 60079-11 / IEC 60079-14 Ex-i standards.			
4. SPARE CAPACITY: All 4 panels maintain >25% spare DIN rail space and spare card slots.			
5. WIRE DUCT: Halogen-free slotted PVC wire duct with snap-on covers (60x80mm & 80x100mm).			

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
4-PANEL SUITE - INTERNAL GENERAL ARRANGEMENT (DOORS OPEN) MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4		DOC NO: CA1-DWG-PL-002 REV: 3.6	DATE: 2026-09-07
SCALE: 1:15 PROJECT: JET COOKER	DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED	SHEET: 2 / 7

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PANEL M1 - RACK C1 HARDWARE ALLOCATION & FIELD TERMINAL SCHEDULE							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-L950TPSXT	ControlLogix 5580 Controller (50MB Memory)	CPU	1	ETH1/2	Plant Master Controller / SCADA	ACTIVE (MASTER)
02	Reserve	Reserved for 1756-EN2TR / Redundancy	RES	-	-	Reserved for Controller Redundancy	SPARE (RESERVE)
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S03-X1,X2	JB-401 Field Signals & Spares	ACTIVE
04	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S04-X1,X2	JB-401 Field Signals	ACTIVE
05	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S05-X1,X2	JB-601 Field Signals	ACTIVE
06	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S06-X1,X2	JB-601 Field Signals & CA1 Spares	ACTIVE
07	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C1S07-X1,X2	JB-401 Solenoid Valves & DO	ACTIVE
08	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C1S08-X1,X2	JB-601 Solenoids & CA1 Spares	ACTIVE
09	1756-IF16	16-Point High Resolution Analog Input	AI	16	C1S09-X1	JB-401 Analog Transmitters (4-20mA)	ACTIVE
10	1756-IF16	16-Point High Resolution Analog Input	AI	16	C1S10-X1	JB-601 Analog Transmitters (4-20mA)	ACTIVE
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)

PANEL M1 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	
- FIELD DESTINATIONS: Connects to field junction boxes: JB-401, JB-601, CA1.	
- POWER INFEEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.	
- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.	
- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).	
- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M1 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1..M4				DOC NO: CA1-DWG-PL-003 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8	DRAWN: AEC-ENG		STATUS: APPROVED		SHEET: 3 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE				

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